

BC356340.1
AP023357.1:8362722-8374865
Actinoplanes sp. NBRC 14428

CA12161319.1
CAKMG010000071.1:113-940
Serratia fonticola

EAM064741.1
ALCMW010000065.1:1-3054
Salmonella enterica

GF05056.1
BLMW1000183.1:87-2582
Lachnospiraceae bacterium

HCA42810.1
DQNW1000178.1:126-5116
Bacteroides bacterium

HEUS277281.1
DASUB010000415.1:87-11650
Ktedonobacteraceae bacterium

HCY20356.1
DQOV0010000286.1:121-2846
Pseudobacteroides sp.

HRN11643.1
DAQM5101000035.1:293-12073
Bacteroides uniformis

MBA2292725.1
JACDAE010000211.1:1-5777
Ktedonobacteraceae bacterium

MBE0152420.1
VNUJ01000181.1:2642-12828
Serratia fonticola

MR4494020.1
JACU0010000374.1:1-7044
Chloroflexi bacterium

MBL865915.1
JAEU1P010000298.1:811-6719
Planctomycetaceae bacterium

MCB9765667.1
JACKEV010000023.1:49688-70136
Alphaproteobacteria bacterium

MCC9663199.1
JADU0101000045.1:1-3337
Polyangiaceae bacterium

MCZ520754.1
JAZ728H10000086.1:1234126-1268533
Verruocisspora sp. MMW2121

MOB8743725.1
JAMQV010000082.1:61384-75986
Ruminococcus bicirculans (ex Wegman et al. 2014)

MDW8244741.1
JAVNM0010000097.1:1-6761
Thermogemata sp.

NEC84092.1
JAGL0101000008.1:199933-22109
Streptomyces sp. SID12581

NEN57038.1
JAMOKC10000156.1:3269-14813
Paenibacillus elgii

OCG47498.1
LZHW1000062.1:389-9760
Gilliamella sp. Choc5-1

OUR88737.1
MAAL01000194.1:32047-45509
Gammaproteobacteria bacterium 42_54_718

PYCO1837.1
QJW010000096.1:115950-135042
Pseudomonas koreensis

WP_001250802.1
NZ_AHAM010000074.1:77849-105342
Cystobacter fuscus DSM 2262

WP_00505287.1
NZ_BK8Z010000087.1:126162-159556
Bacteroides finegoldii CL09703C10

WP_20625492.1
NZ_KB987209.1:686381-704755
Actinonadura aramentaria DSM 43919

WP_27250844.1
NZ_JAAB0101000006.1:1-8398
Serratia fonticola

WP_078837479.1
NZ_JAHF0101000026.1:22215-51711
Streptomyces gottesmaniae

WP_211264672.1
NZ_CP087142.1:2770276-2779437
Gymnella sunshinyii YC6258

WP_200050966.1
NZ_JACVTV010000087.1:46983-63493
Pseudomonas sp. 544

WP_198270198.1
NZ_F0VL01000024.1:9643-21341
Serinonella peptonophila

WP_244912044.1
NZ_WPF401000008.1:369365-377339
Burkholderia aemmonitica

WP_157290296.1
NZ_BAAAZB010000010.1:715511-732322
Chitinophaga oryziarum

WP_089910008.1
NZ_F0B801000002.1:1139288-1161743
Chitinophaga rups

WP_312150096.1
NZ_DALV0010000003.1:315481-136077
Lettliottia nimpresaurilis

WP_2003155131.1
NZ_BPS01000158.1:134-11057
Streptomyces sp. MAC05292

WP_106910914.1
NZ_BQJC010000084.1:153808-171154
Streptomyces sp. MAC07149

WP_129020205.1
NZ_S41001000001.1:961077-970693
Edaphocella flava

WP_130154420.1
NZ_SF0B10000012.1:127-16773
Candidatus Fimmiella inopinata

WP_203204037.1
NZ_SC0B01000133.1:1-1849
Chitinivorax sp. B

WP_138490556.1
NZ_SMPF010000086.1:22889-38222
Dyadobacter ducti

WP_19740344.1
NZ_CP036281.1:5566436-5596820
Polystyrenella longa

WP_106970910.1
NZ_FFKW01000010.1:1-20845
Chitinophaga sp. CF418

WP_16385324.1
NZ_CP046078.1:5169988-5189270
Spartivicius ruber

WP_167016179.1
NZ_VNM001000001.1:2088790-2107550
Chitinophaga sp. Cy-1792

WP_206116207.1
NZ_JAFR01010000026.1:20815-58318
Ruminococcus sp.

WP_170248040.1
NZ_JACBJA010000080.1:15726-21915
Serratia fonticola

WP_100006070.1
NZ_BM0501000001.1:553599-573182
Saccharosporillum sahsuginis

WP_203900227.1
NZ_BONY010000082.1:59094-86759
Rhizocella hellebori

WP_206728872.1
NZ_CP071890.1:5732982-5756908
Pysidicoccus parhayensis

WP_214105476.1
NZ_JAKSC1010000012.1:173370-193621
Acetocarpuspora catenulata

WP_248744487.1
NZ_JALTEA010000016.1:1-9290
Pseudomonas sp. MMJ12-3688

WP_270023004.1
NZ_JAPDP010000001.1:114099-138168
Solirubrobacter phylloclaeae

WP_330421230.1
NZ_JAXCLA010000001.1:554548-577866
Roseateles agri

WS12845.1
CP188362.1:1136087-1162195
Streptomyces sp. NBC_01342

? PNP0X_N ROK ADH_N GGDEF EAL Signa70_r2 Signa70_r3 Signa70_r4 Glyoxalase_6 PT-HIN ? Phage_integrase ? HTH_31 TPRs Pput2613-dean Imn10 rve_3 rve

RHS

? ? SymE_toxin ? SymE_toxin ?

? ? ? ? ?

? ? PT-HINT Tox-PL-2 ? ? Tox-PL-2

RHS_repeat TEN_YD-shell PT-HIN HTH_Tnp_1 LRR_14 LRRs LRR_14 LRRs TIR Cnd1 Ta7_TnSA-like_N rve Mu-transpos_C AAA_22 ? Beta-prop_EML_2 Beta-prop_THOC3

? ? DED0_Tnp_IS110 Transposase_20 ? ?

TssD Phage_GPD Phage_base_V ? ? TssC DUF6531 TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell RHS_repeat TEN_YD-shell ? FRG

PT-HINT Imn30 ? ? RHS_repeat

DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? DUF4253 DUF7820 GnrSD_N MTE5_1575 Vsr DNA_meth_N DNA_methylase

TEN_YD-shell ? TEN_YD-shell TEN_YD-shell TEN_YD-shell DUF3969 PmbA_Tld0_3rd PmbA_Tld0_2nd PmbA_Tld0_1st ABC_tran TOBE_2

? PT-HINT ? ? ? PT-HIN

XhoI XhoI TPRs TPRs ? Aldo_ket_red Cytochrome_CBBs Cu-binding_MopE Cu-binding_MopE Cu-binding_MopE FG-GAP FG-GAP FG-GAP FG-GAP Lipase_GDSL_2 SUKH-3 ? ? SUKH-3 ? ? ?

RHS_repeat ? rve_3 QUF559 ?

SnpB ? TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N DUF179 Laminin Carboxypeptid_reg Carboxypeptid_reg Carboxypeptid_reg Carboxypeptid_reg NID0 RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell ? RHS_repeat TEN_YD-shell TEN_YD-shell ? ? F5_F8_type_C_2 Discoidin CBM_6 Big PKD_4 GSDH ThUA Carboxypeptid_reg Trypsin adh_short Cupin_6 HTH_18

GDP_Manh_Dehyd Epimerase dTPP_sugar_isom NUDIX GATase_6 SIS SIS Van2 DUF6547 PT-HIN ? Phage_integrase Phage_int_SAM_1 ? YjC2 ? Transglutaminase DUF7675

Gly-zipper_Omp TEN_YD-shell MafB19-dean Imn1 PT-HIN ? ? PT-HINT ? PT-HINT GpA_nuclease

AMP-binding AMP-binding_C His_Kinase MEDS STAS_2 STAS ANTAR PemK_toxin ? FG-GAP_3 FG-GAP_3 PT-HIN Phage_integrase Phage_int_SAM_3 HTH_17 RepSA GntR DUF7310 Gp40 HTH_31 HTH_31

TEN_YD-shell ? RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell PT-HIN YeeB YeeA5_toxin ? HTH_26 Resolvase Recombinase ? DED0_Tnp_IS110 Transposase_20

DUF4123 DUF6531 TEN_YD-shell TEN_YD-shell TEN_YD-shell TEN_YD-shell RHS ? ? ? RHS Tox-REase-3 Imn7

TPRs TPRs ? Bac_export_3 ? ? ? ? ? Polypeptide_cyc2 ? Arabinose_bd HTH_18 AcetylCoA_hyd_C CHASES HAMP PAMP GGDEF SBP_bac_3

Catalase RoxA-like Cyt-c Cytochrom_C FKBP_N FKBP_C DUF1232 HTH_3 ? Sel1 DUF6531 TEN_YD-shell TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell ? GDEF CBS EAL QUF3340 Peptidase_S41 PD2 TSP_NTD ADH_N ADH_zinc_N HAD TOBE_2 ABC_tran BPD_transp_1

SWIM SNF2-rel_dom Helicase_C ? ? Imn50 FA_hydroxylase HKD HKD ? FG-GAP_3 FG-GAP_3 RHS_repeat TEN_YD-shell PT-HIN ? OMP_b-brl_4 OMP_b-brl_4 Plug Carboxypeptid_reg ? ? TPRs TPRs DnaJ HSP70 PatA_N STY-kinase

TssD Phage_GPD Phage_base_V ? DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell SM11_KNR4 Josephin DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell SM11_KNR4 Josephin DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell DUF8309 ? Peptidase_S24 TssD KAP_NTase

Peptidase_M48 CTP_transf_like Rsf9 His_Phos_1 Laminin Laminin ? ? SpvB RHS_repeat DUF6443 RHS_repeat TEN_YD-shell Hom_end_hint PT-HIN ? ? PT-HINT RNase_A_bac ? Beta-Lactanase PCHT ?

RHS ? ? ? ACPS 4PPT_N His_Phos_2 Na_sulph_symp CttG

? DUF6247 AMP-binding_C AMP-binding Thiolase_N Thiolase_C_1 Zf-ChsH2 OB_ChH2_C ECH_1 Laminin Laminin Laminin RHS_repeat TEN_YD-shell PT-HIN ? ? RHS_repeat RHS_repeat TEN_YD-shell PT-HINT ? Acyl_transf_3 ? Beta-prop_ACSF4 PQ0_2 STY-kinase

? ? RHS ? RHS Ntox47 ? DUF772 RHS ? RHS ? DUF2947 ? ? ?

EAL CBS GGDEF ? HTH_31 DUF1232 FKBP_C FKBP_N LysR_substrate HTH_1 DcR PAAR_motif DUF6531 RHS_repeat RHS_repeat TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell ? ? ? Imn42 Rieske_2 NIR_SIR NIR_SIR_ferr Fer2_BPD Rubredoxin_C Pyr_redox_2

? NLP_C_P60 ? ? Bro-N ? Peptidase_M23 SLT_2 PT-TG PT-HIN ? ? Toxin-deaminase ? ? Toxin-deaminase ?

OAO ? DUF4154 CHASE8 HAMP GGDEF OmpA Phage_GPD Phage_base_V Gp5_trimer_C DcR PAAR_motif DUF6531 RHS_repeat RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell ? ? RHS_repeat TEN_YD-shell RHS_repeat RHS TreTu_C TriTu GiY-GiY DUF4259

Sigma70_r2 FeCr_C Baculo_1E_1 ? ? ? ? DUF6443 ? ? Asparaginase_2 Mamf_Msf VisAB YiaAB Transglutaminase ?

HTH_18 MPI_cupin_dom His_Kinase His_Kinase ? ? ? RHS_repeat DUF643 ? DUF8198 CHU_C SprB SprB SprB SprB SprB SprB SprB DUF2157 ? ? Methyltransf_24

VipB_2 VipB T6SS_VipA T6SS_HCP ? DUF4225 Phage_GPD Phage_base_V Gp5_trimer_C Gp5_trimer_C DUF2169 Tox-PAAR-like DUF6531 TEN_YD-shell TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat RHS ? SymE_toxin ? ? ? Extensin_1 Autotransporter DUF2230

? T7SS_signal DUF6531 TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell RHS_repeat RHS_repeat RHS_repeat RHS ? Imn58 ? DUF6415 ? ? ?

SP15_N Prim-Pol ABC_tran ? DUF6789 ? ? T7SS_signal DUF6531 RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell ? ? ? ? MDMP1_N HTH_18 D3_1_PfpI ADH_N ADH_zinc_N_2

? ? ? ? ? ? ? ? ? Imn35 ? ? ? ? ?

BON BON HSP20 Bac_DNA_binding ? Bac_DNA_binding ? ? ? SM11_KNR4 ? ChIA ?

TEN_YD-shell ?

DUF772 ECH_1 PAP2 N-OB tRNA-synt_2 DUF6443 TEN_YD-shell ? ? ? ? HTH_Tnp_1 HTH_21 rve rve_3 ? ? Beta-lactanase

DUF1254 DUF1214 DUF1254 DUF1214 DUF1254 DUF1214 ? ? HTH_3 Phage_int_SAM_5 Phage_integrase B1g B1g_13 B1g B1g_13 B1g B1g B1g B1g B1g B1g CADHERIN B1g CADHERIN B1g CFAP143 PT-HIN ? ? Resolvase Recombinase HTH_Tnp_Tc3_2 rve_3 RHS_repeat TEN_YD-shell RHS_repeat ? T6SS_TleI-like_cat T6SS_TleI-like_cat

Lactanase_B Hx18 Snao1_2 ? cNRPB0 adh_short_C2 RHS_repeat DUF5977 DUF6443 ? Pps ? ? ? ? ? QDE_Tnp_1_6

Rn10_sub_bin0 OmpW ? MCPsinal HAMP Ammonium_transp Methyltransf_23 Phage_GPD Phage_base_V Gp5_trimer_C DUF4123 DUF6531 RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? Beta-prop_IFT122_1st Beta-prop_THOC3 Beta-prop_MM02_C WD40_WDHD1_1st Arg SBP_bac_3 BPD_transp_1 N_BPD_transp_1

OTCase_N OTCase AA_kinase Peptidase_M20 Lyase_1 ? ? RHS_repeat DUF6443 ? DUF4169 DUF7710 ? Macro ? ?

CotH Rrf2 Acylphosphatase ACPS AASDHPPPT_N ? Beta_helix Pe19A-like_beta_helix Dockerin_1 Glyco_hydro_18 Cu-oxidase_4 Cytochrom_B562 Transglutaminase PT-HIN ? DUF7385 DUF697 MPR_HSR1 ChII AAA_25 Zn_ribbon_LapB ? ? ?

T6SS_TssF Phage_GPD Phage_base_V Gp5_trimer_C DUF2345 DUF4123 ? ? DUF6531 TEN_YD-shell RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? ? ? ? RHS_repeat RHS ? Htx43

TGT Oueosine_syth Amidohydro_1 DUF5677 ? Para-Soj-MTPase ? DUF6531 RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell RHS_repeat RHS_repeat TEN_YD-shell ? ? DUF1939 MarR_2 GbaR_C Hydrolase SBP_bac_3 BPD_transp_1 N_BPD_transp_1

DUF222 HH dCMP_cyt_deam_1 MmgE_PrpD_C MmgE_PrpD_N Cyclase CoA_transf_3 Urease_alpha Amidohydro_3 AMP-binding_C AMP-binding DUF6531 Ricin_B_lectin Laminin Laminin RHS_repeat RHS_repeat RHS_repeat Laminin FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 RHS_repeat RHS_repeat RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell TEN_YD-shell PT-HINT PT-HIN ? ? TPRs TPR_Mult TPRs AAA_16 BTAD Trans_reg_C Pps DinB_2 Acetylase Peptidase_M23 ? Arginase

ExD0 cF-dsKA_trAr DksA-Like_N G1DA ? Trypsin ? Phage_int_SAM_1 Phage_integrase SpvB FG-GAP FG-GAP_3 FG-GAP_3 FG-GAP_3 TcdB_toxin_n10N RHS_repeat RHS_repeat TEN_YD-shell ? Protease AAA AAA_2 ClpB_D2-small Anisotr_n_3 DnaJ TPRs HSP70 TPRs

Hydrolase Multi_Drug_Res DUF952 DUF7168 DUF2786 ? Ribosomal_L12 MTP SpvB RHS_repeat RHS_repeat Laminin TEN_YD-shell ? ADH_Fe_C Fe-ADH Aldehdh TOBE_2 OB_MaIk ABC_tran TOBE_2 OB_MaIk ABC_tran BPD_transp_1 BPD_transp_1

DUF4123 DUF6531 TEN_YD-shell TEN_YD-shell TEN_YD-shell TEN_YD-shell TEN_YD-shell ? ? ? ? ? ? ?

? ? Sigma70_r4_2 Sigma70_r2 ? ? Peptidase_M1 CsbD RHS_repeat RHS_repeat RHS_repeat PT-HIN ? Alk_phosphatase Hb_5 CHASE2 Vgb_lyase Vgb_lyase TPP_enzyme_C TPP_enzyme_M TPP_enzyme_N Ferritin Abhydrolase_3

Abhydrolase_3 Ala_recarnase_N D_ser_dehydrat Amidase ? ? HTH_Tnp_1 SpvB FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 TcdB_toxin_n10N HYDIN_VesB CFAB65-Like_Ig ASH RHS_repeat RHS_repeat TEN_YD-shell TEN_YD-shell ? ? ? ? TPRs TPRs STN Secretin_N Secretin Cohesin N_methyl_N_methyl T2SS6 ?

? ? SIS HTH_6 SpoIIIE MASE1 R0D HKD HKD Ricin_B_lectin Laminin FG-GAP_3 FG-GAP_3 FG-GAP_3 FG-GAP_3 RHS_repeat TEN_YD-shell TEN_YD-shell PT-HIN ? BDAH_Leukar ? Esterase ? BraA-like Abhydrolase_4 Abhydrolase_1